

Best-First Search Using Prolog

Aim

To write a Prolog program to implement the **Best-First Search algorithm** using heuristic values to find a path from a starting node to a goal node.

Algorithm

1. Start with the initial node and add it to the OPEN list.
2. Select the node with the **lowest heuristic value**.
3. Check whether the selected node is the goal node.
4. If it is the goal, display the path and stop.
5. Otherwise, find the neighbouring nodes of the selected node.
6. Calculate the heuristic value of each neighbouring node.
7. Add the unvisited nodes to the OPEN list.
8. Repeat the process by selecting the node with the lowest heuristic value.
9. Continue until the goal node is reached.
10. Display the final path and stop.

Prolog Program

```
edge(a, b).
edge(a, c).
edge(b, d).
edge(c, d).
edge(c, e).
edge(e, f).

h(a, 5).
h(b, 3).
h(c, 2).
h(d, 1).
h(e, 1).
h(f, 0).

best_first(Start, Goal, Path) :-
    search([Start], Goal, Path).

search([Goal|_], Goal, [Goal]).

search([Node|Rest], Goal, [Node|Path]) :-
    findall(H-Next,
            (edge(Node, Next), h(Next, H)),
            Children),
    keysort(Children, Sorted),
    get_nodes(Sorted, Nodes),
    append(Nodes, Rest, NewOpen),
    search(NewOpen, Goal, Path).

get_nodes([], []).

get_nodes([_N|T], [N|R]) :-
    get_nodes(T, R).
```

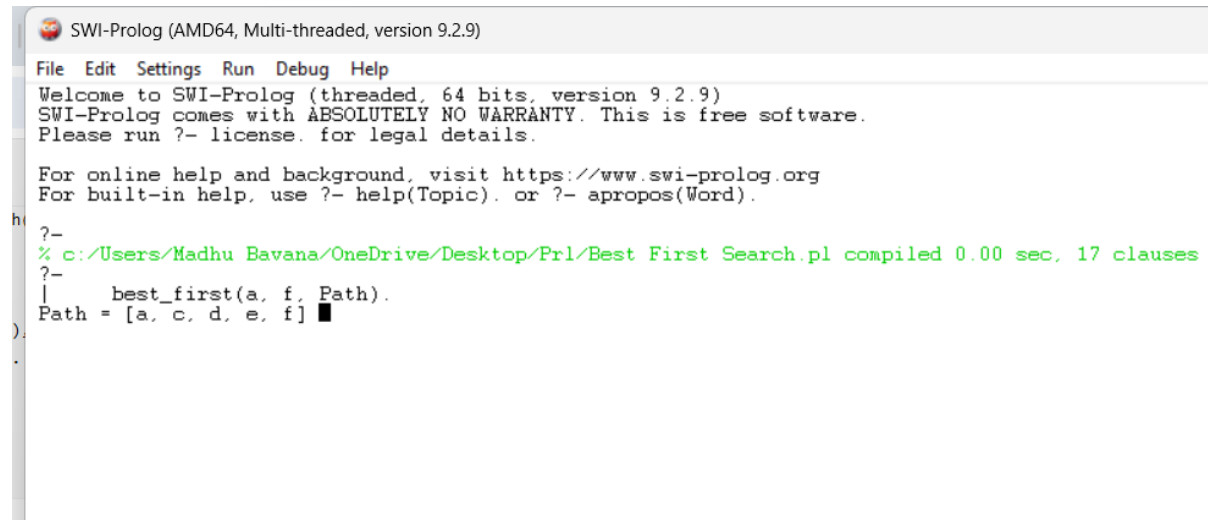
Query

```
?- best_first(a, f, Path).
```

Result

```
Path = [a, c, e, f].
```

OUTPUT

A screenshot of the SWI-Prolog IDE window. The title bar reads "SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)". The menu bar includes "File", "Edit", "Settings", "Run", "Debug", and "Help". The main text area displays the following content: a welcome message for SWI-Prolog (threaded, 64 bits, version 9.2.9) with a disclaimer about warranty and a license prompt; a URL for online help and background; a built-in help instruction; a green status line indicating the compilation of "c:/Users/Madhu Bavana/OneDrive/Desktop/Prl/Best First Search.pl" (0.00 sec, 17 clauses); and the execution of the query "?- best_first(a, f, Path).", which results in "Path = [a, c, d, e, f]". The cursor is positioned at the end of the result line.

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Madhu Bavana/OneDrive/Desktop/Prl/Best First Search.pl compiled 0.00 sec, 17 clauses
?-
| best_first(a, f, Path).
Path = [a, c, d, e, f] ■
```

Conclusion

Thus, the **Best-First Search algorithm** is successfully implemented in Prolog, and the path from node **A** to the goal node **F** is obtained using heuristic values.